Jing Yao

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EDUCATION

Dalian University of Technology **Dalian, China**

Master in Instrument Science and Technology Sept. 2023- Jun. 2026

- Supervisor: Prof. Donghan Ma
- Research Direction: Super-Resolution Microscopy
- Awards: First-Class Academic Scholarship (×3)

Beijing University of Chemical Technology **Beijing, China**

B.S.E. in Robotics Sept. 2020- Jun. 2023

- Relevant Coursework: LinearAlgebra (100), Probability Theory and Mathematical Statistics (99), General Physics (95), Signals and Systems (95), C++ (A), Python (A), MATLAB (94), etc. | GPA: 3.67/4.33 (TOP10%)
- Awards: Li Wen – YangYan Excellent Scholarship, Outstanding Student of Beijing University of Chemical Technology (×2), People’s Scholarship (×2)

B.S.E in Bioengineering Sept. 2018- Jun. 2020

- Relevant Coursework: Organic Chemistry (99), Fundamental Chemistry (96), Physical Chemistry (91), etc. | RANK: 2/60

PUBLICATION

PAPER

- ***Jing Yao***, Jiarui Zhang, Ming Xu, Can Wang, Donghan Ma. Digital-micromirror-device-based illumination strategies for background suppression in single-molecule localization microscopy [J]. **Biomedical Optics Express**.

PATENT

- Donghan Ma, ***Jing Yao***, Jiarui Zhang, Ming Xu, Can Wang; A Background Suppression Method for Single-Molecule Localization Microscopy Based on a Digital Micromirror Device, China.

PROJECT EXPERIENCE

Construction of the Drift Correction Module in the STORM System **Oct. 2024- Feb. 2025**

- Developed the LabVIEW software for system control. (LabVIEW)
- Resolved the issue preventing MATLAB from being invoked and enabled MATLAB calls from LabVIEW for cross-correlation computation. (MATLAB)
- Integrated a shutter into the hardware system and implemented serial communication control. (Serial Communication)
- Conducted biological experiments for validation.

Background Suppression in Deep-tissue Super-resolution Imaging **Sept. 2023- Jun. 2026**

- Designed and built the required optical setup. (SOLIDWORKS, CAD)
- Developed the related image processing algorithms and system registration algorithms. (MATLAB)
- Constructed an integrated hardware – software timing-trigger module. (LabVIEW)
- Conducted biological experiments for validation.

Expression, purification, and crystallization of anti-CRISPR system proteins **Jan. 2020- Dec. 2020**

- Supervisor: Prof. Yue Feng
- Performed core wet-lab experiments, including protein purification, column chromatography, and cell culture/seeding.

SKILLS

- Technical Skills: LabVIEW, MATLAB, C++, SOLIDWORKS, CAD, PLC, etc.
- Languages: Chinese (Native), English (CET-6)